MD. ASAFUDDAULA SOBAHANI SHOUNAK

Product Architect & Engineering Lead
Medical Simulation · Digital Twins · XR (VR/AR) · ECS/DOTS · Simulation Authoring & Workflow Automation

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PROFESSIONAL SUMMARY

Product Architect and Engineering Lead with 5+ years of professional experience delivering high-performance Unity applications across medical simulation, digital twins, and XR training systems. Expert in Unity ECS (DOTS), data-oriented design, and simulation authoring & workflow automation. Proven ability to lead engineering teams, architect scalable systems, and ship enterprise healthcare solutions and consumer products with millions of users.

CORE TECHNICAL SKILLS

UNITY / XR / SIMULATION

Unity Engine (URP, XR Toolkit) · Unity ECS / DOTS · Addressables · Simulation Authoring · Workflow Automation · Medical Simulation · VR / AR (Meta Quest 2/3) · Digital Twins · Multiplayer Systems

PROGRAMMING / BACKEND / TOOLS

C# / .NET / ASP.NET Core · JavaScript · REST APIs · xAPI / LRS / LMS · IIS Deployment · Android Build Management · Performance Optimization · Git / Jira / Agile / Scrum

PROFESSIONAL EXPERIENCE

Product Architect & Engineering Lead

June 2024 – Present

Talent Pro

- Lead Unity development for enterprise-grade medical simulation and XR training platforms.
- Architect ECS-driven systems for scalable scenario execution, assessment logic, and performance-critical workflows.
- Design simulation authoring and workflow automation tools enabling non-technical SMEs to create training scenarios.
- Develop VR medical training applications using Meta Quest 3.
- Integrate xAPI and Learning Record Stores (LRS) for learner analytics and performance tracking.
- Mentor Unity engineers and define engineering standards.

Senior Unity Developer

February 2022 – May 2024

Ghost Interactive

- Shipped Unity products with a combined 1M+ downloads.
- Designed ECS-inspired architectures for gameplay and simulation systems.

- Built internal development tools to improve content production and QA efficiency.
- Optimized rendering, physics, and memory usage across mobile and PC platforms.

Junior Software Engineer (Game)

March 2021 – January 2022

Toontech (Games4Life)

Junior Game Developer

September 2020 – February 2021

Technomagic Pvt Ltd

KEY PROJECTS

SIT CPREP – Clinical Preparedness E-Learning Program

Singapore Institute of Technology

- Developed real-world medical simulation scenarios for radiographers and physiotherapists.
- Implemented ECS-inspired architectures for scenario flow, assessment, and state management.
- Built simulation authoring and workflow automation tools to support rapid content creation.
- Integrated LMS and Learning Record Store (xAPI) for analytics-driven healthcare training.

JLG Crane – Digital Twin (VR)

- Developed a VR-based digital twin enabling real-time crane operation and training.
- Implemented system-driven simulation logic for interaction, physics coordination, and safety validation.

Bus Simulator BD (BSBD) — 2M+ Downloads

- Optimized vehicle physics and rendering for low-end Android devices.
- Contributed to scalable gameplay systems and long-term performance stability.

Trench Warfare — 1M+ Downloads | The Rem — 500K+ Downloads | AlexanderBall: A Countryball Tale (Steam)

- WW1 strategy game with AI-driven gameplay systems; horror survival with modular architecture; large-scale simulation with optimized pathfinding.

EDUCATION

Bachelor of Science (B.Sc.) in Computer Science & Engineering

Jahangirnagar University, Bangladesh CGPA: 3.17 / 4.00

CERTIFICATIONS

UNIVERSITY & CORE GAME DEVELOPMENT (COURSERA)

- Game Design and Development Specialization – Michigan State University *XU38U23UKCJ7*
- Game Design and Development Capstone – Michigan State University *6JK5C8D2GZJE*
- Game Development for Modern Platforms – Michigan State University *29R5UU7GYP9K*
- Business of Games and Entrepreneurship – Michigan State University *FF3ZL86FX83H*
- Principles of Game Design – Michigan State University *PYETUSBA38N7*

- Introduction to Game Development – Michigan State University *VDRKD7ZZETCQ*
- Introduction to C# Programming and Unity – University of Colorado *HJPFVFPQYFBU*
- More C# Programming and Unity – University of Colorado *CFHP556LERX8*

UNITY / XR / AR

- Unity and C# Basics – Meta *KHFYXRKUGYL3*
- Unity Essentials – Unity
- Unity: Building VR User Interfaces – LinkedIn Learning
- Unity: Integrating GPS and Points of Interest – LinkedIn Learning
- AR Development Techniques 01: Basic Concepts – LinkedIn Learning
- AR Development Techniques 02: Lighting and Physics – LinkedIn Learning
- Unity Medical Visualization – LinkedIn Learning
- Unity: Interactivity for AEC – LinkedIn Learning

AI / GENERATIVE AI

- Build Your Generative AI Productivity Skills – Microsoft & LinkedIn
- Claude Code in Action – Anthropic
- Introduction to Generative AI – Google *8336900*

AGILE / SOFTWARE PRACTICES

- Agile Foundations – LinkedIn Learning
- Scrum: The Basics – LinkedIn Learning
- Agile Project Management with Jira Cloud (Projects, Boards, Processes)

INDUSTRY & GOVERNMENT

- Intel Edge AI Scholarship – OpenVINO Fundamentals – Intel Corporation
- Training for Mobile Application Developer (iOS) – ICT Division, Bangladesh *L1A-B-30-0-37*

I hereby declare that all the information provided above is true, complete, and accurate to the best of my knowledge and belief.